

Hard Clustering Problems

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(I) Consider a set of unlabeled data points:

$$\mathcal{X} = \{ \underline{x}_1, \underline{x}_2, \dots, \underline{x}_N \} \text{ where } \underline{x}_i \in \mathbb{R}^d, \forall i \in [N].$$

Clustering provides a partitioning of the original dataset into K clusters such that:

$$\mathcal{X} = \bigcup_{j=1}^K C_j \text{ where}$$

$$C_r \cap C_m = \emptyset, \forall r \neq m$$

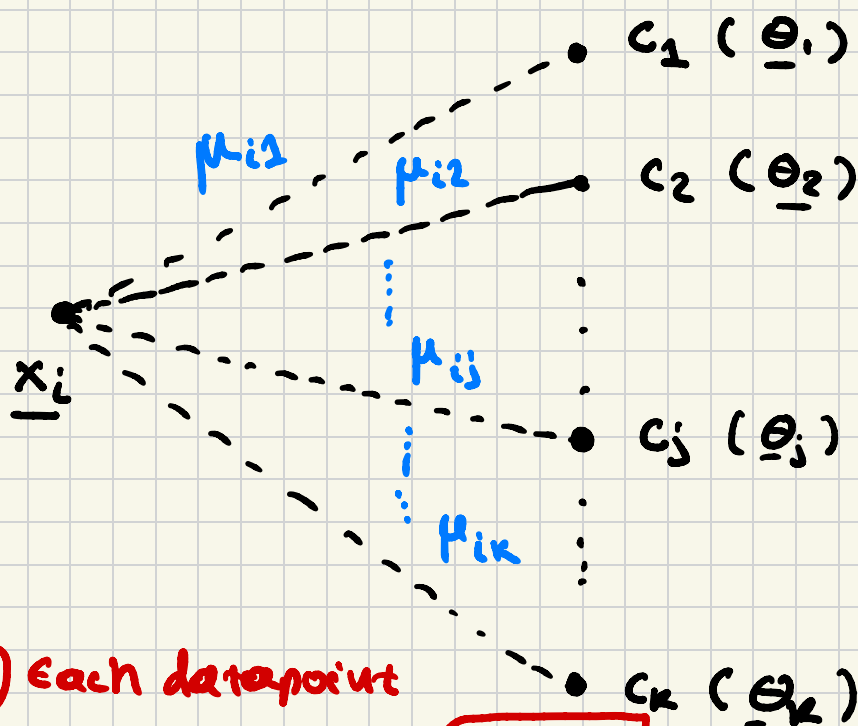
(II) Problem Definition:

Assign each datapoint $\underline{x}_i$ to a unique cluster C_j such that the data points assigned to the j -th cluster exhibit minimum distance relative to the cluster representative $\underline{\theta}_j$.

Memo: Total number of clusters K .

Each cluster is represented by a vector $\underline{\theta}_j$.

(III) Let $\mu_{ij} \in \{0, 1\}$ denote the membership status of the i -th datapoint relative to the j -th cluster $\{1 \leq i \leq N, 1 \leq j \leq K\}$



(*) Each datapoint is assigned to a single cluster.

$$\sum_{j=1}^K \mu_{ij} = 1, \forall i \in [N]$$

Memo : Consider that all membership values are organized in a membership matrix $M = [\mu_{ij}]$ such that $1 \leq i \leq N$ and $1 \leq j \leq K$.

(IV) In this setting, the clustering problem may be defined as an optimization problem where the objective/cost function is given as:

$$\min_{(\underline{\Theta}, \underline{M})} J(\underline{\Theta}; \underline{M}) = \sum_{i=1}^N \sum_{j=1}^K \mu_{ij} \cdot \|\underline{x}_i - \underline{\theta}_j\|^2$$

$$\text{s.t.} \quad \sum_{j=1}^K \mu_{ij} = 1, \quad \forall j \in [K] \quad \boxed{\text{OP. I}}$$

where $\underline{\Theta} = [\underline{\theta}_1, \underline{\theta}_2, \dots, \underline{\theta}_K]$

S.O.S : The objective function is not differentiable since it's not continuous given that $\mu_{ij} \in [0, 1]$.

Memo: This is a hard mixed-integer combinatorial optimization problem since the number of possible clusterings of N datapoints into K clusters will be given by the so-called Stirling numbers of the second type:

$$S(N, K) = \frac{1}{K!} \sum_{j=0}^{K-1} (-1)^{K-j} \binom{K}{j} j^N$$

Therefore, we cannot use straight-forward optimization techniques.

VI Heuristic Optimization Scheme:

(a): Consider that $\underline{\theta}_j$'s with $1 \leq j \leq K$ are fixed.

Since for each vector $\underline{x}_i$ only one μ_{ij} is 1 and all the others are set to 0, it is straight-forward to deduce that $J(\underline{\theta}, \underline{\mu})$ is minimized if we assign each $\underline{x}_i$ to the closest cluster-representative:

$$\mu_{ij} = \begin{cases} 1, & j = \arg \min_{r \in [K]} \|\underline{x}_i - \underline{\theta}_r\|^2; \\ 0, & \text{otherwise,} \end{cases} \quad [A]$$

$\forall i \in [N]$

(β): Consider that μ_{ij} 's are fixed with $1 \leq i \leq N$ and $1 \leq j \leq K$.

In this setting, we need to evaluate the F.O.C, with respect to the parameters $\underline{\theta}_r$, $\forall r \in [K]$, as:

$$\frac{\partial J}{\partial \underline{\theta}_r} = \underline{0} \in \mathbb{R}^p \quad (1) \quad \Leftarrow$$

$$\frac{\partial J}{\partial \underline{\theta}_r} = \frac{\partial}{\partial \underline{\theta}_r} \left[\sum_{i=1}^N \sum_{j=1}^K \mu_{ij} \cdot \|\underline{x}_i - \underline{\theta}_j\|^2 \right] = \underline{0} \quad (2)$$

s.o.s: Since we are differentiating w.r.t. $\underline{\theta}_r$, all the other $\underline{\theta}_m$'s with $m \neq r$ are treated as constants as they do not contribute to the optimization process. They are constant in the context of the partial differentiation process.

Thus, we may write that:

$$\begin{aligned}\frac{\partial J}{\partial \underline{\theta}_r} &= \sum_{i=1}^N \sum_{j=1}^K \frac{\partial}{\partial \underline{\theta}_r} [\mu_{ij} \cdot \|\underline{x}_i - \underline{\theta}_j\|^2] = \\ &= \sum_{i=1}^N \frac{\partial}{\partial \underline{\theta}_r} [\mu_{ir} \|\underline{x}_i - \underline{\theta}_r\|^2] = \underline{\mathbf{0}}, \\ &\quad \forall r \in [K].\end{aligned}$$

⊛ This is equivalent to:

$$\begin{aligned}\sum_{i=1}^N \mu_{ir} \cdot \frac{\partial}{\partial \underline{\theta}_r} [\|\underline{x}_i - \underline{\theta}_r\|^2] &= \underline{\mathbf{0}} \Rightarrow \\ \sum_{i=1}^N \mu_{ir} (-2 \underline{x}_i + 2 \underline{\theta}_r) &= \underline{\mathbf{0}}, \forall r \in [K].\end{aligned}$$

$$\boxed{\frac{\partial J}{\partial \underline{\theta}_r} = \sum_{i=1}^N \mu_{ir} (-2 \underline{x}_i + 2 \underline{\theta}_r) = \underline{\mathbf{0}}, \quad \forall r \in [K]} \quad (3)$$

Memo: $\|\underline{x}_i - \underline{\theta}_r\|^2 = (\underline{x}_i - \underline{\theta}_r)^T (\underline{x}_i - \underline{\theta}_r) =$

$$= \underline{x}_i^T \underline{x}_i - \underline{x}_i^T \underline{\theta}_r - \underline{\theta}_r^T \underline{x}_i + \underline{\theta}_r^T \underline{\theta}_r =$$

$$= \underline{x}_i^T \underline{x}_i - 2 \underline{x}_i^T \underline{\theta}_r + \underline{\theta}_r^T \underline{\theta}_r$$

(i): $\frac{\partial}{\partial \underline{\theta}_r} [\underline{x}_i^T \underline{x}_i - 2 \underline{x}_i^T \underline{\theta}_r + \underline{\theta}_r^T \underline{\theta}_r] =$

$$- 2 \underline{x}_i + 2 \underline{\theta}_r$$

(ii) $\frac{\partial \underline{x}^T \underline{x}}{\partial \underline{x}} = \frac{\partial \underline{x}^T \underline{I} \underline{x}}{\partial \underline{x}} = (\underline{I} + \underline{I}^T) \cdot \underline{x} =$

$$= 2 \underline{I} \underline{x} = 2 \underline{x}$$

From Eq. (3), we may write that:

$$- 2 \sum_{i=1}^N \mu_{ir} \underline{x}_i + 2 \sum_{i=1}^N \mu_{ir} \underline{\theta}_r = \underline{0}, \forall r \in [R]$$

$\Leftrightarrow$

$$\underline{\theta}_r \cdot \sum_{i=1}^N \mu_{ir} = \sum_{i=1}^N \mu_{ir} \underline{x}_i \Leftrightarrow$$

$$\underline{\theta}_r = \frac{\sum_{i=1}^N \mu_{ir} \underline{x}_r}{\sum_{i=1}^N \mu_{ir}}, \quad \forall r \in [K] \cup [B]$$

Eqs. (A) and (B) can pave the way in order to formulate the Generalized Hard Clustering Scheme.

Generalized Hard Clustering Scheme:

Step 1: Choose $\underline{\theta}_j(0)$ as initial estimates for $\underline{\theta}_j$, $\forall j \in [K]$

Step 2: $t=0$

Step 3: Repeat:

Step 3.1: $\forall i \in [N]$: Update μ_{ij} according to:

$$\mu_{ij}(t) = \begin{cases} 1, j = \arg \min_{r \in [K]} \| \underline{x}_i - \underline{\theta}_r(t) \|^2; \\ \emptyset, \text{ otherwise} \end{cases} \quad (c)$$

Step 3.2: $t = t + 1$

Step 3.3: $\forall j \in [K]$: Update $\underline{\theta}_j$ according to:

$$\underline{\theta}_j(t) = \frac{\sum_{i=1}^N \mu_{ij}(t-1) \underline{x}_i}{\sum_{i=1}^N \mu_{ij}(t-1)} \quad (d)$$

Until $\| \underline{\theta}(t) - \underline{\theta}(t-1) \| < \epsilon$

Memo: We may consider that cluster centers are organized as:

$$\underline{\underline{\Theta}} = [\underline{\Theta}_1, \underline{\Theta}_2, \dots, \underline{\Theta}_K] \quad \text{and}$$

$$\underline{\underline{\mu}} = \begin{bmatrix} \underline{\mu}_1 \\ \underline{\mu}_2 \\ \vdots \\ \underline{\mu}_N \end{bmatrix} \quad \text{where}$$

$\underline{\Theta}_j$ is the j -th cluster representative
and

$\underline{\mu}_i$ is the i -th datapoint membership vector.

K-Means ++ Initialization Algorithm:

Step 1: Choose the first cluster representative randomly from the set of the available datapoints.

Step 2: For each subsequent centroid, the probability of selecting each datapoint $\underline{x} \in \pi$ as the next centroid is proportional to its squared Euclidean distance from the nearest already chosen centroid.

Step 3: Repeat until K centroids are chosen.

K-MEANS ++ INITIALIZATION ALGORITHM

$m = 1$

$\underline{\theta}_m(0) = \underline{x}_{i_m}$, where i_m is uniformly sampled from $\{1, 2, \dots, N\}$.

FOR $m = 2$ TO K :

$\underline{\theta}_m(0) = \underline{x}_{i_m}$, where i_m is sampled from $\{1, 2, \dots, N\}$ with probability:

$$p(i_m) = \frac{\|\underline{x}_{i_m} - \underline{\theta}^*\|^2}{\sum_{i=1}^N \|\underline{x}_i - \underline{\theta}^*\|^2}$$

with $\underline{\theta}^* = \min_{j \in [m-1]} \|\underline{x}_{i_m} - \underline{\theta}_j(0)\|^2$